

Grade 7
Unit 8
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

Two students evaluated the expression $\frac{(5^2 \cdot 2.8^0)^4}{5^7}$.

Patricia's Work	Kim's Work	Word Bank
$\frac{(5^2 \cdot 2.8^0)^4}{5^7}$	$\frac{(5^2 \cdot 2.8^0)^4}{5^7}$	zero
$\frac{(5^2 \cdot 1)^4}{5^7}$	$\frac{5^8 \cdot 2.8^0}{5^7}$	identity
$\frac{(5^2)^4}{5^7}$	$\frac{5^8 \cdot 1}{5^7}$	power of a power
$\frac{5^8}{5^7}$	$\frac{5^8}{5^7}$	product of powers
5^1	5^1	power of a product
5	5	quotient of powers
		power of a quotient

Complete the statements to explain each student's work by using the phrases from the word bank for questions 1-6.

- Patricia used the zero exponent law first by rewriting 2.8^0 as 1.
- Next, she multiplied by 1 then she used the power of a power exponent law by multiplying the exponents 4 and 2.
- Patricia then used the quotient of a power exponent law by subtracting 8 and 7. Lastly, she found the value of the expression.
- Kim used the power of a power exponent law first by multiplying 4 and 2 and by multiplying 4 and 0.
- Next, she used the zero exponent law by rewriting 2.8^0 as 1.
- Kim then used the quotient of a power exponent law by subtracting 8 and 7. Finally, she found the value of the exponential expression.

Use the laws of exponents to evaluate the expressions.

$$7. \quad \left(\frac{1.5^5}{1.5^3}\right)^2 \quad (1.5^{5-3})^2 = (1.5^2)^2 = 1.5^4 \\ 1.5 \times 1.5 \times 1.5 \times 1.5 \\ 5.0625$$

$$8. \quad \left[\left(\frac{4}{5}\right)^0 \cdot \left(\frac{4}{5}\right)^1\right]^2 \quad \left(1 \cdot \frac{4}{5}\right)^2 = \left(\frac{4}{5}\right)^2 = \frac{4^2}{5^2} = \frac{16}{25}$$

$$9. \quad 4.7^2 \cdot (4.7^1)^3 \quad 4.7^2 \cdot 4.7^3 = 4.7^{2+3} = 4.7^5 = 2,293.45007$$

$$10. \quad \frac{9.9}{8.8^0} \cdot 9.9^3 \quad \frac{9.9^1}{1} \cdot 9.9^3 = 9.9^4 = 9.9 \times 9.9 \times 9.9 \times 9.9 = 9605.9601$$